

Revised Generation Interconnection System Impact Study Report

For Queue Project Q15-0477

Thomas Hill Station 500 kV
January 03, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0477. The Interconnection Customer proposes a 922.5 MW natural gas combined cycle facility interconnecting at the Thomas Hill Station 500 kV substation in the Indiana area. The request was received on May 15, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 228 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$48,092,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: CARLEY ROAD to JAMES DE YOUNG 230kV
Contingency: E METHUEN 500/230kV XFMR
Loading: 121.5% (Rating: 463 MVA)
DFAX: 10.0%
Pre-project loading: 101.5%
2. Facility: MUSKOGEE to DEAN FOREST 230kV
Contingency: SWAN VLY to MUSKOGEE 500kV line
Loading: 113.4% (Rating: 506 MVA)
DFAX: 7.3%
Pre-project loading: 100.1%
3. Facility: DEAN FOREST to EASTOWN RD 230kV
Contingency: SWAN VLY SUBSTATION to MUSKOGEE 500kV
Loading: 112.2% (Rating: 498 MVA)
DFAX: 7.3%
Pre-project loading: 98.7%
4. Facility: E METHUEN to GRAY'S FLAT 230kV
Contingency: E METHUEN to MADELINE 230kV line
Loading: 109.2% (Rating: 644 MVA)
DFAX: 13.3%
Pre-project loading: 90.1%
5. Facility: ST. LAWRENCE to JAMES DE YOUNG 230kV L
Contingency: LONETREE to E METHUEN 500kV Line
Loading: 108.4% (Rating: 823 MVA)
DFAX: 16.0%
Pre-project loading: 90.5%

6. Facility: MUSKOGEE 500-230 kV TX
Contingency: SWAN VALLEY to MUSKOGEE 500kV
Loading: 106.4% (Rating: 959 MVA)
DFAX: 14.1%
Pre-project loading: 92.9%
7. Facility: BURRIS PIT to HEBO 230kV
Contingency: TREGO to BEAVER 500kV line
Loading: 103.9% (Rating: 650 MVA)
DFAX: 10.5%
Pre-project loading: 89.1%
8. Facility: BURRIS PIT to JAMES DE YOUNG 230kV
Contingency: TREGO to BEAVER 500kV Line
Loading: 103.9% (Rating: 671 MVA)
DFAX: 10.5%
Pre-project loading: 89.5%
9. Facility: MADELIE SUBSTATION to EAST METHUEN SUBSTATION 230kV
Contingency: BARAGA VLG to GRAY'S FLAT 230kV
Loading: 103.5% (Rating: 903 MVA)
DFAX: 14.1%
Pre-project loading: 89.1%
10. Facility: ADIRONDACK to JAMES DE YOUNG 230kV line
Contingency: JAMES DE YOUNG to ST. LAWRENCE 230kV
Loading: 103.1% (Rating: 631 MVA)
DFAX: 9.7%
Pre-project loading: 88.9%
11. Facility: BARAGA VLG to GRAY'S FLAT 230kV
Contingency: MADELINE to E METHUEN 230kV
Loading: 101.6% (Rating: 874 MVA)
DFAX: 13.3%
Pre-project loading: 87.5%
12. Facility: E METUEN 500/230 kV Transformer
Contingency: BEAVER to TREGO 500kV
Loading: 100.1% (Rating: 1,394 MVA)
DFAX: 21.5%
Pre-project loading: 85.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CARLEY ROAD to JAMES DE YOUNG 230kV	Rebuild line with higher-capacity conductor	\$23,755,000	10% (\$2,411,000)
DEAN FOREST to MUSKOGEE 230kV line	Rebuild line with higher-capacity conductor	\$53,005,000	22% (\$11,888,000)
DEAN FOEST to EASTOWN RD 230kV	Rebuild line with higher-capacity conductor	\$21,648,000	22% (\$4,855,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
GRAY'S FLAT to EAST METUEN 230kV	Reconductor line	\$15,762,000	10% (\$1,623,000)
ST. LAWRENCE to JAMES DE YOUNG 230kV Line	Reconductor line	\$21,369,000	15% (\$3,272,000)
MUSKOGEE 500/230 kV XFMR	Replace transformer with higher-rated unit	\$42,798,000	37% (\$15,794,000)
BURRIS PIT to HEBO 230kV	Reconductor line	\$8,436,000	10% (\$881,000)
BURRIS PIT to JAMES DE YOUNG 230kV	Reconductor line	\$20,905,000	10% (\$2,184,000)
E METHUEN to MADELINE 230kV line	Reconductor line	\$9,849,000	10% (\$1,037,000)
ADIRONDACK to JAMES DE YOUNG 230kV line	Reconductor line	\$10,907,000	10% (\$1,141,000)
GRAY'S FLAT to BARAGA VLG 230kV Line	Reconductor line	\$7,573,000	10% (\$763,000)
E METUEN 500/230 kV Transformer	Replace transformer with higher-rated unit	\$22,254,000	10% (\$2,243,000)
	Total allocated to project		\$48,092,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.